

Efficient ML — Final Project Proposal

Stress-Testing Wanda (Sun et al., ICLR 2024)

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Paper Information

- **Title:** A Simple and Effective Pruning Approach for Large Language Models
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- **Code:** github.com/locuslab/wanda **Track:** Track 2 (Stress-Testing)

Paper Summary

Wanda scores each weight by $S_{ij} = |W_{ij}| \cdot \|X_j\|_2$ —the product of weight magnitude and input activation norm—computed from 128 C4 calibration samples. Weights are compared per output neuron and the lowest-scoring are removed. This requires no weight updates or gradient computation, making it up to $300\times$ faster than SparseGPT while matching its WikiText-2 perplexity at 50% unstructured sparsity across LLaMA models. Performance collapses above 60% sparsity.

Why This Paper is Meaningful and Suspicious

Meaningful. Wanda’s insight—that activation magnitude matters as much as weight magnitude for pruning—is grounded in the outlier feature phenomenon (Dettmers et al., 2022) and has become the dominant LLM pruning baseline with over 650 citations. Its speed and simplicity make it practical for deployment.

Suspicious. The activation norm $\|X_j\|_2$ is a statistical fingerprint of the calibration data. The paper claims robustness to calibration data, but tests this only by varying the *quantity* of English C4 samples, never the *domain or language*. Since features specific to Korean are dormant on English text, English calibration assigns near-zero importance scores to them and marks them for removal. Several recent papers support this: calibration source shifts pruning quality (Ji et al., ICLR 2025), standard Wanda degrades multilingual performance even at 50% sparsity (Choenni et al., EMNLP 2025), and calibration-dependent pruning can overfit its calibration distribution (Shin et al., EMNLP 2024).

Yet no existing paper directly tests: **can wrong-domain calibration make Wanda perform worse than calibration-free magnitude pruning?** M-Wanda used mixed multilingual calibration and cross-lingual NLU tasks, not pure single-language calibration on Korean-native benchmarks. Ji et al. tested only English calibration and evaluation. No one has compared domain-mismatched Wanda against the calibration-free magnitude baseline to check whether using calibration data can be actively counterproductive.

Methodology

We test a single thesis: **Wanda’s calibration data distribution determines which model capabilities survive pruning, and domain-mismatched calibration can reverse Wanda’s advantage over the calibration-free magnitude pruning baseline.** All experiments use LLaMA, which scores above random chance on Korean benchmarks (KMMLU, KoBEST), with three baselines: dense (upper bound), magnitude pruning (calibration-free), and SparseGPT with C4 (calibration-dependent).

Experiment 1 — Calibration domain shift at 50% sparsity. Prune with four calibration sources: C4 English (default), MC4-ko (Korean), The Stack (code), and random tokens (control). Evaluate on WikiText-2 perplexity, MMLU (English knowledge), GSM8K (math reasoning), and KoBEST HellaSwag (Korean language). We predict English calibration will damage Korean capabilities more than Korean calibration damages English, because Korean text naturally contains some English but English text contains no Korean. We specifically test whether English-calibrated Wanda on KoBEST HellaSwag falls to or below magnitude pruning.

Experiment 2 — Per-task fragility curves across sparsity. Prune with C4 at 50%, 60%, and 70% sparsity. Evaluate on the same four benchmarks. We expect WikiText-2 perplexity to degrade continuously while GSM8K and KoBEST HellaSwag cliff-drop at different thresholds, which lets us identify which capabilities break first and at what sparsity.

Experiment 3 — Calibration $\times$ sparsity interaction. We test the extreme corner: English calibration and Korean calibration both at 70% sparsity, evaluated on all four benchmarks. If the Korean-capability loss from English calibration (Experiment 1) and from high sparsity (Experiment 2) compound super-additively, this would suggest both effects target the same low-activation features.

Total: ~ 26 pruning runs on $8\times$ NVIDIA RTX 6000 Ada GPUs (48 GB each). LLaMA-2-7B supplements with English-only analysis to test whether cross-lingual shift produces more severe damage than within-English domain shift.